

Міністерство освіти і науки України
Національний Університет “Львівська Політехніка”



Лабораторна робота
з дисципліни
“Об'єктно-орієнтоване програмування, частина 2”

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Лабораторна робота 4.

Тема: Використання SQL для роботи з даними.

Мета: Навчитися виконувати SQL-запити, використовуючи SQLite та Pandas для роботи з даними.

Завдання:

```
import pandas as pd
import sqlite3
import io
```

```
# =====
```

1. Завантаження CSV

```
# =====
```

```
try:
```

```
    from google.colab import files
    print("Виберіть файл Job_opportunities.csv для завантаження:")
    uploaded = files.upload()
    first_file = list(uploaded.keys())[0]
    df = pd.read_csv(io.BytesIO(uploaded[first_file]))
    print(f'Файл '{first_file}' завантажено!")
```

```
except ImportError:
```

```
    # Якщо запуск не в Colab, намагаємось прочитати локально
    df = pd.read_csv("Job_opportunities.csv")
    print("Файл Job_opportunities.csv завантажено локально!")
```

```
print(f'Рядків: {len(df)}, Стовпців: {len(df.columns)} ")
print(df.head())
```

```
# =====
```

2. Обробка зарплат для аналітики

```
# =====
```

```
# Витягуємо числа з рядка типу "£50,000 - £70,000"
```

```
df["Min Salary"] = df["Salary Range"].str.extract(r"£([\d,]+)")[0].str.replace(",", "", "").astype(int)
```

```
df["Max Salary"] = df["Salary Range"].str.extract(r"£([\d,]+\s*\s*£([\d,]+)")[0].str.replace(",", "", "").astype(int)
```

```
df["Avg Salary"] = (df["Min Salary"] + df["Max Salary"]) // 2
```

```
# =====
```

3. Підключення до SQLite

```
# =====
```

```
conn = sqlite3.connect(':memory:')
df.to_sql("jobs", conn, index=False, if_exists="replace")
print("Дані завантажено у таблицю 'jobs' у SQLite.\n")
```

```
# =====
```

4. Основні запити SQL

```
# =====
```

```
def run_query(title, query):
    print(f"\n--- {title} ---")
    result = pd.read_sql(query, conn)
    print(result.to_string(index=False))
    return result
```

```
run_query("Перші 10 вакансій", "SELECT * FROM jobs LIMIT 10;")
```

```
run_query("Вакансії з Required Skills SQL",
          "SELECT * FROM jobs WHERE `Required Skills` LIKE '%SQL%';")
```

```
run_query("Унікальні міста", "SELECT DISTINCT Location FROM jobs ORDER
BY Location;")
```

```
run_query("Унікальні компанії", "SELECT DISTINCT Company FROM jobs
ORDER BY Company;")
```

```
# =====
```

5. Аналітичні запити

```
# =====
```

```
run_query("Середня зарплата за Experience Level",
          "SELECT `Experience Level`, ROUND(AVG(`Avg Salary`), 0) AS
avg_salary "
          "FROM jobs GROUP BY `Experience Level`;")
```

```
run_query("Кількість вакансій за Experience Level",
          "SELECT `Experience Level`, COUNT(*) AS vacancy_count "
          "FROM jobs GROUP BY `Experience Level`;")
```

```
run_query("Мінімальна та максимальна зарплата",
          "SELECT MIN(`Min Salary`) AS min_salary, MAX(`Max Salary`) AS
max_salary, "
```

```

"ROUND(AVG(`Avg Salary`), 0) AS avg_salary FROM jobs;")

# =====
# 6. Агрегатні запити
# =====
run_query("Вакансії > £50,000 за Industry",
    "SELECT Industry, COUNT(*) AS count FROM jobs WHERE `Min Salary`
> 50000 "
    "GROUP BY Industry ORDER BY count DESC;")

run_query("Середня зарплата за Industry",
    "SELECT Industry, ROUND(AVG(`Avg Salary`), 0) AS avg_salary "
    "FROM jobs GROUP BY Industry ORDER BY avg_salary DESC;")

# =====
# 7. Складніші запити
# =====
run_query("Вакансії за Location та Experience Level",
    "SELECT Location, `Experience Level`, COUNT(*) AS count "
    "FROM jobs GROUP BY Location, `Experience Level` ORDER BY
Location;")

run_query("Вакансії за Industry та Job Type",
    "SELECT Industry, `Job Type`, COUNT(*) AS count "
    "FROM jobs GROUP BY Industry, `Job Type` ORDER BY Industry;")

# =====
# 8. Додаткові запити
# =====
run_query("ТОП-5 вакансій за Max Salary",
    "SELECT `Job Title`, Company, Location, `Max Salary` "
    "FROM jobs ORDER BY `Max Salary` DESC LIMIT 5;")

# Аналіз навичок через Pandas
skills_series = df["Required Skills"].str.split(",").explode().str.strip()
skill_counts = skills_series.value_counts().head(15)
print("\n--- ТОП-15 Навичок за кількістю вакансій ---")
print(skill_counts)

```

```
# =====
# 9. Закриття з'єднання
# =====

conn.close()
print("\nЗ'єднання з базою даних закрито.")
```

Результат:

...  Виберіть файл Job opportunities.csv для завантаження:

Вибрати файли

Job opportunities.csv

Job opportunities.csv(text/csv) - 56360 bytes, last modified: 01.03.2026 - 100% done

Saving Job opportunities.csv to Job opportunities (6).csv

Файл 'Job opportunities (6).csv' завантажено!

Рядків: 364, Столпців: 10

	Job Title	Job Description	Required Skills	\
0	Software Engineer	Develop software	Java, Python	
1	Data Analyst	Analyze data	SQL, Excel	
2	Network Engineer	Maintain networks	Cisco, WAN	
3	Cloud Architect	Design cloud	AWS, Azure	
4	Cybersecurity Analyst	Protect data	Cybersecurity	

	Salary Range	Location	Company	Experience Level	\
0	£40,000 - £60,000	London	ABC Tech	Entry-Level	
1	£35,000 - £50,000	Manchester	XYZ Analytics	Junior	
2	£45,000 - £70,000	Birmingham	Network Solutions	Mid-Level	
3	£60,000 - £90,000	Edinburgh	Cloud Innovators	Senior	
4	£50,000 - £80,000	Glasgow	SecureGuard	Mid-Level	

	Industry	Job Type	Date Posted
0	Technology	Full-Time	1/5/2023
1	Analytics	Full-Time	2/10/2023
2	Networking	Full-Time	3/15/2023
3	Cloud Computing	Full-Time	4/20/2023
4	Security	Full-Time	5/25/2023

Дані завантажено у таблицю 'jobs' у SQLite.

--- Перші 10 вакансій ---

Job Title	Job Description	Required Skills	Salary Range	Location	Company	Experience Level	Industry	Job Type	Date Posted	Min Salary	Max Salary
Software Engineer	Develop software	Java, Python	£40,000 - £60,000	London	ABC Tech	Entry-Level	Technology	Full-Time	1/5/2023	40000	60000
Data Analyst	Analyze data	SQL, Excel	£35,000 - £50,000	Manchester	XYZ Analytics	Junior	Analytics	Full-Time	2/10/2023	35000	50000
Network Engineer	Maintain networks	Cisco, WAN	£45,000 - £70,000	Birmingham	Network Solutions	Mid-Level	Networking	Full-Time	3/15/2023	45000	70000
Cloud Architect	Design cloud	AWS, Azure	£60,000 - £90,000	Edinburgh	Cloud Innovators	Senior	Cloud Computing	Full-Time	4/20/2023	60000	90000
Cybersecurity Analyst	Protect data	Cybersecurity	£50,000 - £80,000	Glasgow	SecureGuard	Mid-Level	Security	Full-Time	5/25/2023	50000	80000
IT Project Manager	Manage projects	PMP, Agile	£70,000 - £100,000	Leeds	Project Leaders	Senior	Project Management	Full-Time	6/30/2023	70000	100000
Data Scientist	Analyze big data	R, Python	£55,000 - £85,000	Bristol	Data Insights	Mid-Level	Data Science	Full-Time	7/5/2023	55000	85000
DevOps Engineer	Streamline devops	Docker, Jenkins	£50,000 - £75,000	Newcastle	DevOps Solutions	Mid-Level	DevOps	Full-Time	8/10/2023	50000	75000
IT Support Analyst	Provide IT support	Windows, Linux	£30,000 - £45,000	Sheffield	IT Services	Entry-Level	Support	Full-Time	9/15/2023	30000	45000
UX/UI Designer	Design user interfaces	UI/UX Design	£40,000 - £70,000	Cardiff	Design Studios	Junior	Design	Full-Time	10/20/2023	40000	70000

--- Ваканції з Required Skills SQL ---

Job Title	Job Description	Required Skills	Salary Range	Location	Company	Experience Level	Industry	Job Type	Date Posted	Min Salary
Data Analyst	Analyze data	SQL, Excel	£35,000 - £50,000	Manchester	XYZ Analytics	Junior	Analytics	Full-Time	2/10/2023	35000
Database Analyst	Manage databases	SQL, Oracle, Database Management	£40,000 - £65,000	London	Data Solutions	Mid-Level	Database	Full-Time	11/25/2023	40000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£50,000 - £85,000	Edinburgh	Data Engineers	Mid-Level	Data Engineering	Full-Time	12/10/2023	50000
Data Analyst Intern	Analyze data as an intern	SQL, Excel	£20,000 - £30,000	Birmingham	Data Interns	Internship	Analytics	Full-Time	1/15/2023	20000
Data Engineer	Build data pipelines	Big Data, Python, SQL	£55,000 - £85,000	Cardiff	Data Innovators	Mid-Level	Data Engineering	Full-Time	1/5/2023	55000
Database Analyst	Manage databases	SQL, Oracle, Database Management	£40,000 - £65,000	Manchester	Data Solutions	Mid-Level	Database	Full-Time	11/25/2023	40000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£50,000 - £85,000	Cardiff	Data Engineers	Mid-Level	Data Engineering	Full-Time	1/5/2022	50000
Data Analyst	Analyze data	SQL, Data Visualization	£40,000 - £65,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	2/10/2022	40000
Database Analyst	Manage databases	SQL, Oracle, Database Management	£40,000 - £65,000	Birmingham	Data Solutions	Mid-Level	Database	Full-Time	11/25/2022	40000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£50,000 - £85,000	Bristol	Data Engineers	Mid-Level	Data Engineering	Full-Time	12/10/2022	50000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£40,000 - £70,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	4/20/2021	40000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£45,000 - £75,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	4/15/2021	45000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£45,000 - £75,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	4/15/2021	45000
Database Administrator	Manage databases	SQL, Database Management	£50,000 - £85,000	Manchester	Data Management	Mid-Level	Database	Full-Time	4/19/2021	50000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£60,000 - £95,000	Bristol	Data Engineers	Mid-Level	Data Engineering	Full-Time	9/9/2021	60000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£45,000 - £75,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	4/15/2021	45000
Database Administrator	Manage databases	SQL, Database Management	£50,000 - £85,000	Manchester	Data Management	Mid-Level	Database	Full-Time	4/19/2021	50000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£60,000 - £95,000	Bristol	Data Engineers	Mid-Level	Data Engineering	Full-Time	9/9/2021	60000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£45,000 - £75,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	3/15/2020	45000
Database Administrator	Manage databases	SQL, Database Management	£50,000 - £85,000	Manchester	Data Management	Mid-Level	Database	Full-Time	3/19/2020	50000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£60,000 - £95,000	Bristol	Data Engineers	Mid-Level	Data Engineering	Full-Time	8/9/2020	60000
Data Analyst	Analyze data trends	SQL, Excel, Data Analysis	£45,000 - £75,000	Manchester	Data Insights	Junior	Data Analysis	Full-Time	3/15/2019	45000
Database Administrator	Manage databases	SQL, Database Management	£50,000 - £85,000	Manchester	Data Management	Mid-Level	Database	Full-Time	3/19/2019	50000
Data Engineer	Build data pipelines	ETL, Big Data, SQL	£60,000 - £95,000	Bristol	Data Engineers	Mid-Level	Data Engineering	Full-Time	8/9/2019	60000

--- Унікальні міста ---

Location
Birmingham
Bristol
Cardiff
Edinburgh
Glasgow
Leeds
Liverpool
London
Manchester
Newcastle
Nottingham
Sheffield

--- Унікальні компанії ---

Company
ABC Tech
AI Ethics
AI Innovations
Asset Managers
Audit Experts
Biz Analysts
Blockchain Tech
Change Managers
Cloud Analysts
Cloud Architects
Cloud Innovations
Cloud Innovators
Cloud Interns
Cloud Migrators
Cloud Security
Cloud Support
CodeCrafters
Compliance Experts
Compliance Specialists
Continuity Managers
Cybersecurity Engineers
Data Engineers
Data Governance
Data Innovators
Data Insights
Data Interns
Data Management
Data Managers
Data Privacy
Data Solutions
Design Studios
Dev Interns
DevOps Solutions
Full-stack Devs
IT Analyst Trainees
IT Analysts
IT Consultancy
IT Consultants
IT Governance Mgmt
IT Helpdesk
IT Insights

...
IT Helpdesk
IT Insights
IT Services
IT Solutions
IT Support Experts
IT Support Specialists
IT Training
IT Training Assist
IT Training Specialists
ML Engineers
Mobile Devs
Network Admin Trainees
Network Experts
Network Solutions
ProcureIT
Procurement Analysts
Procurement Coordinators
Procurement Mgmt
Project Coordinators
Project Leaders
QA Analysts
QA Automation
QA Solutions
Risk Analysts
Sales Leaders
Sales Reps
Secure DevOps
SecureCloud
SecureGuard
SecureIT
SecureNet
Security Consultants
Service Desk
Software Architects
System Services
Tech Innovations
Tech Solutions
UX Designers
UX Innovations
UX Labs
UX Research Assistants
Web Creations
Web Studios
XYZ Analytics


```

*** --- Середня зарплата за Experience Level ---
Experience Level  avg_salary
Entry-Level      37500.0
Internship       25000.0
Junior           54638.0
Mid-Level        70949.0
Senior           93679.0

```

```

--- Кількість вакансій за Experience Level ---
Experience Level  vacancy_count
Entry-Level      17
Internship       3
Junior           58
Mid-Level        216
Senior           70

```

```

*** --- Вакансії > £50,000 за Industry ---
Industry  count
Cloud Computing  22
Security        19
Procurement     11
Project Management  9
Data Science    9
Governance      8
Web Development  7
Software Design  7
Sales           7
Ethics          7
DevOps          7
Data Privacy    7
Consulting      7
Compliance      7
Blockchain      7
Artificial Intelligence  7
Analytics       7
Quality Assurance  5
Networking      5
Data Engineering  5
Mobile Development  4
Business Analysis  4
Audit           4
Machine Learning  1
Change Management  1
Business Continuity  1
Asset Management  1

```

```

--- Средняя зарплата за Industry ---
...
      Industry avg_salary
      Software Design 107500.0
      Governance      93333.0
      Sales            93125.0
      Ethics           90000.0
      Business Continuity 90000.0
      Consulting       87857.0
      Analytics        84167.0
      Mobile Development 81429.0
      Artificial Intelligence 81429.0
      Machine Learning 80000.0
      Change Management 80000.0
      Project Management 79219.0
      Compliance       78750.0
      Cloud Computing  78400.0
      Data Science     78333.0
      Blockchain       78214.0
      Procurement      78194.0
      Security         74821.0
      Data Privacy     74286.0
      Data Engineering 72813.0
      Asset Management 72500.0
      DevOps           71000.0
      Business Analysis 70000.0
      Audit            69286.0
      Risk Management  66429.0
      Networking       63889.0
      Web Development  63295.0
      Software Development 61389.0
      Database         61071.0
      Design           60059.0
      Quality Assurance 58833.0
      Data Analysis    58214.0
      IT Services      57500.0
      IT Analysis      52500.0
      Support          51100.0
      Technology       50000.0
      Education        48500.0
      Research         37500.0
      Training         30000.0
      Cloud            25000.0

```

--- Ваканції за Location та Experience Level ---

```
*** Location Experience Level count
Birmingham Entry-Level 1
Birmingham Internship 2
Birmingham Junior 7
Birmingham Mid-Level 30
Birmingham Senior 8
Bristol Entry-Level 2
Bristol Junior 4
Bristol Mid-Level 16
Bristol Senior 2
Cardiff Entry-Level 2
Cardiff Junior 6
Cardiff Mid-Level 16
Cardiff Senior 5
Edinburgh Entry-Level 2
Edinburgh Junior 2
Edinburgh Mid-Level 15
Edinburgh Senior 16
Glasgow Entry-Level 1
Glasgow Junior 13
Glasgow Mid-Level 23
Glasgow Senior 2
Leeds Entry-Level 1
Leeds Junior 2
Leeds Mid-Level 20
Leeds Senior 12
Liverpool Junior 4
Liverpool Mid-Level 12
London Entry-Level 3
London Internship 1
London Junior 4
London Mid-Level 19
London Senior 1
Manchester Entry-Level 2
Manchester Junior 11
Manchester Mid-Level 20
Manchester Senior 10
Newcastle Entry-Level 2
Newcastle Junior 1
Newcastle Mid-Level 11
Newcastle Senior 1
Nottingham Mid-Level 8
```

```

--- Вакансії за Industry та Job Type ---
***
      Industry Job Type count
Artificial Intelligence Full-Time 7
      Analytics Full-Time 9
      Asset Management Full-Time 1
      Audit Full-Time 7
      Blockchain Full-Time 7
      Business Analysis Full-Time 10
      Business Continuity Full-Time 1
      Change Management Full-Time 1
      Cloud Full-Time 1
      Cloud Computing Full-Time 25
      Compliance Full-Time 8
      Consulting Full-Time 7
      Data Analysis Full-Time 7
      Data Engineering Full-Time 8
      Data Privacy Full-Time 7
      Data Science Full-Time 9
      Database Full-Time 7
      Design Full-Time 17
      DevOps Full-Time 10
      Education Full-Time 15
      Ethics Full-Time 7
      Governance Full-Time 9
      IT Analysis Full-Time 1
      IT Services Full-Time 3
      Machine Learning Full-Time 1
      Mobile Development Full-Time 7
      Networking Full-Time 9
      Procurement Full-Time 18
      Project Management Full-Time 16
      Quality Assurance Full-Time 15
      Research Full-Time 1
      Risk Management Full-Time 7
      Sales Full-Time 8
      Security Full-Time 28
      Software Design Full-Time 7
      Software Development Full-Time 9
      Support Full-Time 30
      Technology Full-Time 1
      Training Full-Time 1
      Web Development Full-Time 22

```

```

--- ТОП-5 вакансій за Max Salary ---
      Job Title      Company      Location      Max Salary
Software Architect Software Architects Nottingham 140000
Software Architect Software Architects Nottingham 140000
Software Architect Software Architects Nottingham 140000
Software Architect Software Architects Nottingham 140000
IT Governance Manager IT Governance Mgmt Birmingham 130000

```

```

--- ТОП-15 Навичок за кількістю вакансій ---

```

```

Required Skills
AWS          32
Azure        32
Python       25
SQL          24
Troubleshooting 23
JavaScript   23
Compliance   23
Windows      19
Training      17
IT Strategy   15
IT Education  14
Procurement  14
UX Design     14
Cloud Analysis 12
Node.js       12
Name: count, dtype: int64

```

```

З'єднання з базою даних закрито.

```

Висновок:

Під час виконання лабораторної роботи було опановано механізм імпорту даних із формату CSV у середовище Pandas з подальшою міграцією у базу даних SQLite. Застосування SQL-запитів, зокрема агрегатних функцій та операцій групування, дозволило систематизувати інформацію про ринок праці за рівнем доходу, галузями та регіонами. Результати роботи підтверджують, що інтеграція інструментів Python із реляційними базами даних забезпечує високу ефективність під час аналізу великих масивів структурованої інформації.